

ФІКТ Гр. ІПЗ-24-2[1]

```

    fuelType: z.enum(['petrol', 'diesel', 'electric', 'hybrid']),
    market: z.enum(['European', 'US', 'Asian']).default('European')
  });

export const updateCarSchema = createCarSchema.partial();

export type CreateCarInput = z.infer<typeof createCarSchema>;

export type CarEntity = CreateCarInput & {
  id: string;
  createdAt: Date;
  updatedAt: Date;
};

```

3.3. Завдання 3: Модуль зберігання (in-memory storage)

Лістинг:

```

import * as crypto from 'crypto';
import { CarEntity, CreateCarInput } from '../schemas/entity.schema';

const carsStorage = new Map<string, CarEntity>();

export const getAllCars = (filters?: { fuelType?: string; year?: number; market?: string }): CarEntity[] => {
  let cars = Array.from(carsStorage.values());

  if (filters?.fuelType) {
    cars = cars.filter(car => car.fuelType === filters.fuelType);
  }
  if (filters?.year) {
    cars = cars.filter(car => car.year === Number(filters.year));
  }
  if (filters?.market) {
    cars = cars.filter(car => car.market === filters.market);
  }

  return cars;
};

export const getCarById = (id: string): CarEntity | undefined => {
  return carsStorage.get(id);
};

export const createCar = (data: CreateCarInput): CarEntity => {
  const id = crypto.randomUUID();
  const now = new Date();

  const newCar: CarEntity = {
    ...data,
    id,
    createdAt: now,
    updatedAt: now,
  };

  carsStorage.set(id, newCar);
  return newCar;
};

export const updateCar = (id: string, data: Partial<CreateCarInput>): CarEntity | null => {

```

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```

const existingCar = carsStorage.get(id);
if (!existingCar) return null;

const updatedCar: CarEntity = {
  ...existingCar,
  ...data,
  updatedAt: new Date(),
};

carsStorage.set(id, updatedCar);
return updatedCar;
};

export const deleteCar = (id: string): boolean => {
  return carsStorage.delete(id);
};

export const resetCars = (): void => {
  carsStorage.clear();
};

```

3.4. Завдання 4: CRUD-маршрути та middleware

Лістинг:

```

import { Router, Request, Response } from 'express';
import { createCar, deleteCar, getAllCars, getCarById, updateCar } from
'../storage/entity';
import { createCarSchema, updateCarSchema } from '../schemas/entity.schema';
import { validate } from '../middleware/validate';

const router = Router();

router.get('/european', (_req: Request, res: Response) => {
  const cars = getAllCars({market: 'European'});
  res.status(200).json(cars);
});

router.get('/', (req: Request, res: Response) => {
  const cars = getAllCars(req.query);
  res.status(200).json(cars);
});

router.get('/:id', (req: Request<{ id: string }>, res: Response) => {
  const car = getCarById(req.params.id);
  if(!car){
    return res.status(404).json({message: 'Car not found'});
  }
  res.status(200).json(car);
});

router.post('/', validate(createCarSchema), (req: Request, res: Response) => {
  const newCar = createCar(req.body);
  res.status(201).json(newCar); //201 created
});

router.patch('/:id', validate(updateCarSchema), (req: Request<{ id: string }>, res:
Response) => {
  const updatedCar = updateCar(req.params.id, req.body);
  if(!updatedCar){
    return res.status(404).json({message: 'Car not found'});
  }
});

```

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```

    }
    res.status(200).json(updatedCar);
  });

router.delete('/:id', (req: Request<{ id: string }>, res: Response) => {
  const isDeleted = deleteCar(req.params.id);
  if (!isDeleted) {
    return res.status(404).json({message: 'Car not found'});
  }
  res.status(204).send(); //204 no content
});

export default router;

```

3.5. Завдання 5: Розділення app.ts та server.ts

app.ts

```

import express = require('express');
import cors = require('cors');
import carRoutes from './routes/entity';
import { errorHandler } from './middleware/errorHandler';

const app = express();

app.use(cors());
app.use(express.json());

app.use('/cars', carRoutes);

app.use(errorHandler);

export default app;

```

server.ts

```

import app from './app';

const PORT = 3000;

app.listen(PORT, () => {
  console.log(`Сервер запущено: http://localhost:${PORT}`);
});

```

3.6. Завдання 6: Інтеграційні тести з Supertest

Лістинг:

```

import request = require('supertest');
import app from '../src/app';
import { resetCars } from '../src/storage/entity';

beforeEach(() => {
  resetCars();
});

describe('Cars API Integration Tests', () => {

  describe('POST /cars', () => {
    it('1. should create a new car with valid data', async () => {

```

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```

    const res = await request(app)
      .post('/cars')
      .send({
        model: 'Toyota Camry',
        year: 2022,
        fuelType: 'petrol',
        market: 'Asian'
      });

    expect(res.status).toBe(201);
    expect(res.body).toHaveProperty('id');
    expect(res.body.model).toBe('Toyota Camry');
    expect(res.body).toHaveProperty('createdAt');
  });

  it('2. should return 400 if validation fails (missing required field)',
  async () => {
    const res = await request(app)
      .post('/cars')
      .send({ year: 2022 });

    expect(res.status).toBe(400);
    expect(res.body).toHaveProperty('errors');
  });

  it('3. should return 400 if year is out of bounds', async () => {
    const res = await request(app)
      .post('/cars')
      .send({ model: 'Ford T', year: 1500, fuelType: 'petrol' });

    expect(res.status).toBe(400);
  });

  describe('GET /cars', () => {
    it('4. should return an empty array if no cars exist', async () => {
      const res = await request(app).get('/cars');
      expect(res.status).toBe(200);
      expect(res.body).toEqual([]);
    });

    it('5. should return all created cars', async () => {
      await request(app).post('/cars').send({ model: 'Car A', year: 2020,
      fuelType: 'petrol' });
      await request(app).post('/cars').send({ model: 'Car B', year: 2021,
      fuelType: 'diesel' });

      const res = await request(app).get('/cars');
      expect(res.status).toBe(200);
      expect(res.body).toHaveLength(2);
    });

    it('6. should filter cars by ALL query parameters', async () => {
      await request(app).post('/cars').send({ model: 'Old US Car', year:
      2000, fuelType: 'petrol', market: 'US' });
      await request(app).post('/cars').send({ model: 'New Euro EV', year:
      2023, fuelType: 'electric', market: 'European' });
      await request(app).post('/cars').send({ model: 'New Asian Petrol',
      year: 2023, fuelType: 'petrol', market: 'Asian' });

      const resYear = await request(app).get('/cars?year=2023');

```

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```

        expect(resYear.status).toBe(200);
        expect(resYear.body).toHaveLength(2);

        const resFuel = await request(app).get('/cars?fuelType=petrol');
        expect(resFuel.body).toHaveLength(2);

        const resMarket = await request(app).get('/cars?market=Asian');
        expect(resMarket.body).toHaveLength(1);
        expect(resMarket.body[0].model).toBe('New Asian Petrol');

        const resCombined = await
request(app).get('/cars?year=2023&fuelType=electric&market=European');
        expect(resCombined.body).toHaveLength(1);
        expect(resCombined.body[0].model).toBe('New Euro EV');
    });
});

describe('GET /cars/european (Special Route)', () => {
    it('7. should return only European market cars', async () => {
        await request(app).post('/cars').send({ model: 'BMW 3', year: 2020,
fuelType: 'petrol', market: 'European' });
        await request(app).post('/cars').send({ model: 'Ford Mustang', year:
2021, fuelType: 'petrol', market: 'US' });

        const res = await request(app).get('/cars/european');
        expect(res.status).toBe(200);
        expect(res.body).toHaveLength(1);
        expect(res.body[0].model).toBe('BMW 3');
    });
});

describe('GET /cars/:id', () => {
    it('8. should return a car by ID', async () => {
        const createRes = await request(app).post('/cars').send({ model:
'Mazda 6', year: 2018, fuelType: 'petrol' });
        const carId = createRes.body.id;

        const getRes = await request(app).get(`/cars/${carId}`);
        expect(getRes.status).toBe(200);
        expect(getRes.body.id).toBe(carId);
    });

    it('9. should return 404 if car is not found', async () => {
        const res = await request(app).get('/cars/non-existent-id');
        expect(res.status).toBe(404);
        expect(res.body.message).toBe('Car not found');
    });
});

describe('PATCH /cars/:id', () => {
    it('10. should update a car successfully', async () => {
        const createRes = await request(app).post('/cars').send({ model:
'Honda Civic', year: 2015, fuelType: 'petrol' });
        const carId = createRes.body.id;

        const patchRes = await request(app)
            .patch(`/cars/${carId}`)
            .send({ year: 2016 });

        expect(patchRes.status).toBe(200);
    });
});

```

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```

        expect(patchRes.body.year).toBe(2016);
        expect(patchRes.body.model).toBe('Honda Civic');
    });

    it('11. should return 400 if update data is invalid', async () => {
        const createRes = await request(app).post('/cars').send({ model: 'Audi A4', year: 2020, fuelType: 'diesel' });

        const patchRes = await request(app)
            .patch(`/cars/${createRes.body.id}`)
            .send({ fuelType: 'water' });

        expect(patchRes.status).toBe(400);
    });

    it('12. should return 404 when trying to update non-existent car', async () => {
        const res = await request(app).patch('/cars/fake-id').send({ model: 'Ghost Car' });
        expect(res.status).toBe(404);
    });

    describe('DELETE /cars/:id', () => {
        it('13. should delete a car and return 204 status', async () => {
            const createRes = await request(app).post('/cars').send({ model: 'Nissan Leaf', year: 2019, fuelType: 'electric' });
            const carId = createRes.body.id;

            const deleteRes = await request(app).delete(`/cars/${carId}`);
            expect(deleteRes.status).toBe(204);

            const getRes = await request(app).get(`/cars/${carId}`);
            expect(getRes.status).toBe(404);
        });

        it('14. should return 404 when trying to delete non-existent car', async () => {
            const res = await request(app).delete('/cars/fake-id');
            expect(res.status).toBe(404);
        });
    });

    describe('Global Error Handler', () => {
        it('15. should handle generic server errors (500)', async () => {
            const res = await request(app)
                .post('/cars')
                .set('Content-Type', 'application/json')
                .send('{"bad": json}');

            expect(res.status).toBe(500);
        });
    });
});

```

3.7. Завдання 7: Фільтрація через query-параметри

Лістинг:

```

export const getAllCars = (filters?: { fuelType?: string; year?: number; market?: string }): CarEntity[] => {
    let cars = Array.from(carsStorage.values());

```

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```

if (filters?.fuelType) {
  cars = cars.filter(car => car.fuelType === filters.fuelType);
}
if (filters?.year) {
  cars = cars.filter(car => car.year === Number(filters.year));
}
if (filters?.market) {
  cars = cars.filter(car => car.market === filters.market);
}

return cars;
};

```

Результат виконання тестів:

PASS lab03/tests/entity.test.ts (5.588 s)

Cars API Integration Tests

POST /cars

- ✓ 1. should create a new car with valid data (103 ms)
- ✓ 2. should return 400 if validation fails (missing required field) (15 ms)
- ✓ 3. should return 400 if year is out of bounds (13 ms)

GET /cars

- ✓ 4. should return an empty array if no cars exist (14 ms)
- ✓ 5. should return all created cars (46 ms)
- ✓ 6. should filter cars by ALL query parameters (61 ms)

GET /cars/european (Special Route)

- ✓ 7. should return only European market cars (20 ms)

GET /cars/:id

- ✓ 8. should return a car by ID (13 ms)
- ✓ 9. should return 404 if car is not found (8 ms)

PATCH /cars/:id

- ✓ 10. should update a car successfully (14 ms)
- ✓ 11. should return 400 if update data is invalid (14 ms)
- ✓ 12. should return 404 when trying to update non-existent car (6 ms)

DELETE /cars/:id

- ✓ 13. should delete a car and return 204 status (19 ms)
- ✓ 14. should return 404 when trying to delete non-existent car (8 ms)

Global Error Handler

- ✓ 15. should handle generic server errors (500) (57 ms)

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File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	100	89.28	100	100	
src	100	100	100	100	
app.ts	100	100	100	100	
src/middleware	100	100	100	100	
errorHandler.ts	100	100	100	100	
validate.ts	100	100	100	100	
src/routes	100	100	100	100	
entity.ts	100	100	100	100	
src/schemas	100	100	100	100	
entity.schema.ts	100	100	100	100	
src/storage	100	85	100	100	
entity.ts	100	85	100	100	9-15
Test Suites: 1 passed, 1 total Tests: 15 passed, 15 total Snapshots: 0 total Time: 6.286 s Ran all test suites matching entity.					

Висновок: набув практичних навичок розробки RESTful API на Express.js з TypeScript та написав інтеграційні тести з використанням Supertest.

Git: <https://git.ztu.edu.ua/ipz/2024-2028/ipz-24-2/Hurynchuk-Sviatoslav/nodejs-labs>

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